

A GOLDEN RATIO-BASED CONSTANT AND CONJECTURES ON ITS POLYLOGARITHMIC VALUES

TRISTEN HARR

ABSTRACT. This paper introduces and analyzes a specific complex constant, denoted Λ_{G1} , which is derived from fundamental properties of the golden ratio, ϕ . The motivation for this constant arises from considering components based on inverse powers of ϕ . We first prove the constant's fundamental properties, including the convergence condition $|\Lambda_{G1}| < 1$. This property allows the constant to be used as an argument in the defining series for the Polylogarithm function, $\text{Li}_s(z)$. We present high-precision numerical evaluations for the Dilogarithm ($s = 2$) and Trilogarithm ($s = 3$) of this constant and, based on this evidence, we conjecture that the values $\text{Li}_s(\Lambda_{G1})$ are transcendental for all integers $s \geq 2$ and do not belong to the field extension $\mathbb{Q}(\pi, e, \ln(2), \phi, \gamma)$.

1. INTRODUCTION

The study of special functions and constants often reveals connections between different areas of mathematics. This paper focuses on the properties of a specific complex constant derived from the golden ratio, $\phi = (1 + \sqrt{5})/2$. The motivation for this constant stems from Euclid's classical problem of dividing a line segment in extreme and mean ratio. We define the components of our constant directly from inverse powers of ϕ , providing a non-arbitrary foundation for its construction.

The purpose of this paper is to rigorously define this constant, Λ_{G1} , and establish its validity as an argument in certain infinite series. This allows us to define and analyze the values of the Polylogarithm function at this specific point. The paper culminates in a formal conjecture about the transcendental nature of these values, based on high-precision numerical analysis.

2. THE CONSTANT Λ_{G1} AND ITS FOUNDATIONAL PROPERTIES

2.1. Defining Components from the Golden Ratio. We define two real constants, T and J , directly from the algebraic properties of the golden ratio.

Definition 2.1 (Primary Real Components). The constants T and J are defined from the golden ratio, ϕ , as:

$$T = \frac{1}{2\phi} \quad \text{and} \quad J = \frac{1}{2\phi^2}$$

These definitions lead to a pair of linear relations, which highlights their connection to classical geometry.

Proposition 2.2 (System of Equations). *The constants T and J are the unique real numbers (a, b) that satisfy the system:*

- (1) $a/b = \phi$.
- (2) $a + b = 1/2$.

Proof. The ratio is immediate from the definitions: $T/J = (1/(2\phi))/(1/(2\phi^2)) = \phi^2/\phi = \phi$. For the sum, we use the identity $\phi + 1 = \phi^2$:

$$T + J = \frac{1}{2\phi} + \frac{1}{2\phi^2} = \frac{\phi + 1}{2\phi^2} = \frac{\phi^2}{2\phi^2} = \frac{1}{2}$$

The system has a unique solution because the determinant of the corresponding linear system in a and b is non-zero. $\square$

Remark. This formulation shows that the constraint $T + J = 1/2$ is not an arbitrary choice, but a direct consequence of defining the components from successive inverse powers of ϕ . As discussed in Appendix A, this corresponds to the classical geometric problem of dividing a line segment of length $L = 1/2$ in the golden ratio. Explicitly, $T = (\sqrt{5} - 1)/4$ and $J = (3 - \sqrt{5})/4$.

From these constants, we define our primary object of study.

Definition 2.3 (The Primary Constant). The constant Λ_{G1} is defined as the complex number $\Lambda_{G1} = T + iJ$.

2.2. Convergence Property. A crucial property of Λ_{G1} is that its magnitude is less than one, which guarantees the convergence of the power series we will investigate.

Lemma 2.4 (Magnitude of Λ_{G1}). *The constant Λ_{G1} lies within the unit disk, i.e., $|\Lambda_{G1}| < 1$.*

Proof. We require $|\Lambda_{G1}|^2 < 1$. We calculate the squared values of the components: $T^2 = \left(\frac{\sqrt{5}-1}{4}\right)^2 = \frac{6-2\sqrt{5}}{16}$ and $J^2 = \left(\frac{3-\sqrt{5}}{4}\right)^2 = \frac{14-6\sqrt{5}}{16}$. Summing these gives:

$$\begin{aligned} |\Lambda_{G1}|^2 &= T^2 + J^2 = \frac{6 - 2\sqrt{5} + 14 - 6\sqrt{5}}{16} \\ &= \frac{20 - 8\sqrt{5}}{16} = \frac{5 - 2\sqrt{5}}{4} \approx 0.131966 \end{aligned}$$

Since $0.131966 < 1$, the condition is satisfied. $\square$

3. EVALUATION OF A RELATED POLYNOMIAL-WEIGHTED SERIES

As a consequence of Lemma 2.4, Λ_{G1} is a valid argument in the well-known series whose sum is given by Eulerian polynomials, $A_k(x)$ [1]. This provides a direct evaluation for a family of series with Λ_{G1} as the base. For any integer $k \geq 0$:

$$\sum_{n=0}^{\infty} n^k \cdot (\Lambda_{G1})^n = \frac{A_k(\Lambda_{G1})}{(1 - \Lambda_{G1})^{k+1}}$$

While this is a direct application of a standard formula, it is included for completeness and context.

4. DEFINING THE POLYLOGARITHM VALUES OF Λ_{G1}

The property $|\Lambda_{G1}| < 1$ is central to this paper, as it allows us to define the Polylogarithm values at this constant. The Polylogarithm function, a key object in this study, is defined by the following power series [2].

Definition 4.1 (The Polylogarithm Function). For any complex s , the Polylogarithm function of order s , $\text{Li}_s(z)$, is defined by:

$$\text{Li}_s(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^s}$$

This series converges for all complex numbers z such that $|z| < 1$.

Since Λ_{G1} satisfies the convergence condition, we can use it as an argument to define the specific numerical values that are the subject of our main conjecture:

$$\text{Li}_s(\Lambda_{G1}) = \sum_{n=1}^{\infty} \frac{(\Lambda_{G1})^n}{n^s}$$

This identification holds for any complex s .

5. NUMERICAL ANALYSIS AND A CONJECTURE ON THE NATURE OF $\text{Li}_s(\Lambda_{G1})$

5.1. Numerical Values. We present high-precision numerical evaluations for the first two non-trivial integer cases of the Polylogarithm of Λ_{G1} . The values were computed using the series definition in Python with the ‘mpmath’ library [3], using a working precision of 200 decimal digits to ensure that any potential integer relations would be detected.

- For $s = 2$, the Dilogarithm of Λ_{G1} is:

$$\text{Li}_2(\Lambda_{G1}) \approx 0.322328253720993 + 0.226730769307749i$$

- For $s = 3$, the Trilogarithm of Λ_{G1} is:

$$\text{Li}_3(\Lambda_{G1}) \approx 0.310345112613931 + 0.198884393433621i$$

5.2. A Conjecture on the Nature of These Values. The transcendental nature of special function values at algebraic arguments is a central topic in number theory. While the values of polylogarithms at roots of unity are often known or conjectured to be expressible in terms of other fundamental constants (e.g., Catalan’s constant), the nature of $\text{Li}_s(z)$ for more general algebraic arguments z , such as Λ_{G1} , is largely unknown. This makes Λ_{G1} a candidate for exploring this challenging landscape.

Our high-precision values for the Dilogarithm ($s = 2$) and Trilogarithm ($s = 3$) could not be resolved into a closed-form expression involving known mathematical constants by integer relation algorithms (like PSLQ). This failure provides numerical evidence for their transcendental nature and leads us to propose the following general conjecture.

Conjecture 5.1. *For any integer $s \geq 2$, the value $\text{Li}_s(\Lambda_{G1})$ is a transcendental number. Furthermore, it is conjectured that this number is not in the field $\mathbb{Q}(\pi, e, \ln(2), \phi, \gamma)$.*

Remark. Proving this conjecture is beyond the scope of this paper, but proposing it frames a clear direction for future work.

6. CONCLUSION

This paper has introduced a complex constant, Λ_{G1} , whose definition is naturally derived from the golden ratio. We have proven that its magnitude is less than one, which validates its use as an argument in the Polylogarithm function. The primary contribution of this work is the introduction of this constant and the subsequent analysis of its polylogarithmic values, $\text{Li}_s(\Lambda_{G1})$.

Based on high-precision numerical evidence from the $s = 2$ and $s = 3$ cases, we have proposed a new open problem for the community: a conjecture on the transcendental nature of these values. The constant’s specific algebraic properties make it a focused case for the broader study of the analytic properties of special functions at algebraic arguments.

7. FUTURE WORK

The conjecture presented in Section 5 opens a clear path for future investigation. A primary goal would be to prove or disprove the transcendental nature of $\text{Li}_s(\Lambda_{G1})$. Further work could also explore other families of series or integral transforms evaluated at Λ_{G1} , or investigate whether the constant’s unique algebraic properties have applications in other areas of mathematics or physics.

APPENDIX A. GEOMETRIC MOTIVATION FOR THE DEFINING EQUATIONS

The definitions of T and J in Section 2 have a direct geometric motivation. They arise from considering the division of a line segment according to the golden ratio. If a segment of length L is divided into two pieces, a and b , such that $a/b = \phi$, then the pieces have lengths $a = L\phi/(\phi + 1)$ and $b = L/(\phi + 1)$. Using the identity $\phi + 1 = \phi^2$, these become $a = L/\phi$ and $b = L/\phi^2$.

Our constants T and J correspond to this construction for the specific case where $L = 1/2$. As shown in Proposition 2.2, this choice of $L = 1/2$ is not an external axiom but rather the natural sum that arises when the components T and J are defined from the fundamental quantities $1/(2\phi)$ and $1/(2\phi^2)$. This provides a non-arbitrary basis for the constant’s definition, rooted entirely in the algebraic properties of ϕ .

REFERENCES

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